

Student Guide — senior / module-09- subtraction-nikhilam

IKS Curriculum

Table of Contents

Student Workbook

Student Workbook — Subtraction — Nikhilam (All from 9) (Senior)

This workbook is yours to write in. Use it for vocab, exercises, and reflection across all 10 days.

Vocabulary tracker

Fill in as we go:

- *Nikhilam Navataścaramam Daśataḥ* — _____
- *pratiyogi (modern coinage)* — _____

Day-by-day notes

Day 1 — The Sūtra — All from 9, Last from 10

What I learned:

One question I have:

Day 2 — Complement to 1000 and Beyond

What I learned:

One question I have:

Day 3 — Subtract from a Power of 10

What I learned:

One question I have:

Day 4 — General Subtraction via Complements

What I learned:

One question I have:

Day 5 — Why It Works — The Algebra

What I learned:

One question I have:

Day 6 — Bar Model Proof

What I learned:

One question I have:

Day 7 — Two's Complement in Computers

What I learned:

One question I have:

Day 8 — Speed Day

What I learned:

One question I have:

Day 9 — Mixed Practice

What I learned:

One question I have:

Day 10 — Final Assessment

What I learned:

One question I have:

Final reflection

After Day 10, write a paragraph: what is the most important thing you learned in this module?

Day 1 — The Sūtra — All from 9, Last from 10

Module: Subtraction — Nikhilam (All from 9) · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: compute the complement of any 2-digit number to 100.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *Nikhilam Navataścaramaṁ Daśataḥ* (IAST: Nikhilam Navataścaramaṁ Daśataḥ; lit. “all from 9, the last from 10”)

Primary source (today’s reading)

From *Bharati Krishna Tirthaji* — see `sources.md`.

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “What’s one thing you remember about yesterday’s lesson?”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Pose the question.** “What’s $100 - 47$?” Most students compute 53 by borrowing. Ask: “What’s $1000 - 437$?” Trickier — the borrowing chains. Today’s method removes the chain entirely.
2. **State the sūtra.** *Nikhilam Navataścaramaṁ Daśataḥ* — “all from 9, the last from 10.” Write the Sanskrit and IAST on the board.
3. **Apply to $100 - 47$.** Subtract each digit of 47 from 9, except the last which is from 10:
 - 4 from 9 = 5
 - 7 from 10 = 3
 - Answer: **53**. No borrowing.
4. **Pattern recognition.** For $1000 - 437$: $9-4=5$, $9-3=6$, $10-7=3 \rightarrow$ **563**. Verify: $437 + 563 = 1000$. ✓
5. **Source note.** Tirthaji’s *Vedic Mathematics* (1965), sūtra #2. Honest history: this is one of the more solidly-rooted methods in older Indian and Arabic arithmetic traditions (decimal

complement was widely used pre-modern). Independent practice: 10 problems on the worksheet.

Activity (15 min)

A scaled version of *Complement Flashcards* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces The Sūtra — All from 9, Last from 10.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Day 2 — Complement to 1000 and Beyond

Module: Subtraction — Nikhīlam (All from 9) · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: compute complements to 1000 and 10000 mentally.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See [lesson-plan.md](#) for any day-specific materials)

Vocabulary introduced today

- *pratiyogi* (modern coinage) (IAST: pratiyogi (modern coinage); lit. “the number that adds to a base”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Define complement.** The complement of N to base B is $B - N$. Today we find complements to 1000 mentally.
2. **Worked examples.**
 - $738 \rightarrow 9-7=2, 9-3=6, 10-8=2 \rightarrow \mathbf{262}$. Check: $738 + 262 = 1000$. ✓
 - $405 \rightarrow 9-4=5, 9-0=9, 10-5=5 \rightarrow \mathbf{595}$. Check: $405 + 595 = 1000$. ✓
 - $999 \rightarrow 0, 0, 1 \rightarrow \mathbf{001}$. Edge case: trailing 9s carry differently — “*all from 9 except the LAST non-zero digit, which is from 10.*”
3. **The “trailing zero” caveat.** For 730 (ends in 0): treat the last non-zero digit as the one from 10. $730 \rightarrow 9-7=2, 10-3=7, 0 \rightarrow \mathbf{270}$. Or equivalently: complement 73 to 100 (= 27), append the trailing 0 $\rightarrow 270$.
4. **Pattern algebra.** For $N < 10^k$, the complement is $(10^k - 1) - N + 1 = (10^k - N)$. The “all from 9 last from 10” procedure computes this digit-by-digit.
5. **Practice.** 15-problem mental sprint, complements to 1000 only. 5 minutes.

Activity (15 min)

A scaled version of *Complement Flashcards* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Complement to 1000 and Beyond.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.

- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Day 3 — Subtract from a Power of 10

Module: Subtraction — Nikhilam (All from 9) · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: compute $1000-567$ and similar instantly.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *Nikhilaṁ Navataścaramaṁ Daśataḥ* (IAST: Nikhilaṁ Navataścaramaṁ Daśataḥ; lit. “all from 9, the last from 10”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **The clean application.** Whenever we’re subtracting from 10^k (10, 100, 1000, 10000, ...), Nikhilam gives the answer directly. No borrowing, no carries.
2. **Speed examples.**
 - $10 - 4 = 6$ (4 from 10)
 - $100 - 37 = 63$ (3 from 9, 7 from 10)
 - $1000 - 268 = 732$
 - $10000 - 1547 = 8453$
3. **Practical use case.** Computing change at a shop. “*You paid ₹1000 for an item costing ₹427. How much change?*” Nikhilam: $9-4=5$, $9-2=7$, $10-7=3 \rightarrow ₹573$. Mental math, no paper.
4. **Verify with traditional subtraction.** Have students cross-check 2 problems by the borrowing method. Same answer. The Nikhilam is just a smarter procedure for the same arithmetic.

5. **Class discussion.** “*When IS Nikhilam genuinely faster than borrowing? When does borrowing keep up?*” (Borrowing keeps up for small numbers like 10–4. Nikhilam dominates for larger powers.)

Activity (15 min)

A scaled version of *Complement Flashcards* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Subtract from a Power of 10.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 4 — General Subtraction via Complements

Module: Subtraction — Nikhilam (All from 9) · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: use add-the-complement trick for arbitrary subtractions.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See [lesson-plan.md](#) for any day-specific materials)

Vocabulary introduced today

- *pratiyogi* (modern coinage) (IAST: *pratiyogi* (modern coinage); lit. “the number that adds to a base”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Beyond powers of 10.** Most real subtractions aren’t from a clean 10^k . So how does Nikhilam help with, say, $6347 - 2589$?
2. **The “add-the-complement” trick.** Rewrite: $6347 - 2589 = 6347 + (10000 - 2589) - 10000 = 6347 + (\text{complement of } 2589) - 10000$.
 - Complement of 2589 to 10000: $9-2=7$, $9-5=4$, $9-8=1$, $10-9=1 \rightarrow 7411$.
 - $6347 + 7411 = 13758$.
 - $13758 - 10000 = 3758$.
 - Check: $6347 - 2589 = 3758$. ✓
3. **When is this faster?** When the borrowing in standard subtraction would chain through many digits, the complement method skips it entirely. For simple subtractions, standard is fine.
4. **Worksheet.** 8 problems, mixed: some clean Nikhilam, some general using add-the-complement.
5. **Reflection.** “*Which method-type did you find genuinely faster? Which felt the same?*”
Honest answers — Nikhilam is a tool, not a universal speed-up.

Activity (15 min)

A scaled version of *Complement Flashcards* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces General Subtraction via Complements.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 5 — Why It Works — The Algebra

Module: Subtraction — *Nikhilam* (All from 9) · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: derive the place-value identity that makes the rule work.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See *lesson-plan.md* for any day-specific materials)

Vocabulary introduced today

- *Nikhilam Navataścaramaṁ Daśataḥ* (IAST: *Nikhilam Navataścaramaṁ Daśataḥ*; lit. “all from 9, the last from 10”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **State the identity.** For N with k digits, $10^k - N$ can be computed as $((10^k - 1) - N) + 1$.
2. **Why “all from 9 last from 10” works.**
 - $(10^k - 1)$ is a number with k nines: 9, 99, 999, 9999, ...
 - $(10^k - 1) - N$: each digit of N is subtracted from 9. The result has no borrows because each digit of $(10^k - 1)$ is 9.
 - Adding 1 to that bumps the last digit by 1, which is equivalent to “the last digit from 10” rather than “from 9.”
3. **Worked algebraic example.** $N = 437$, $k = 3$.
 - $999 - 437 = 562$ (digit-wise: 9-4, 9-3, 9-7)
 - $562 + 1 = 563 = 1000 - 437$. ✓

4. **The general formula.** $10^k - N = (10^k - 1) - N + 1$ — a foundational identity in modular arithmetic and binary computer arithmetic.
5. **Senior bridge.** This is the same identity that underlies *two's complement* in binary computer arithmetic — tomorrow's topic.

Activity (15 min)

A scaled version of *Complement Flashcards* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Why It Works — The Algebra.

Wrap-up (5 min)

- One-sentence exit ticket: “*What's one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today's task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 6 — Bar Model Proof

Module: Subtraction — Nikhilam (All from 9) · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: draw a visual proof using bar models and number lines.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See `lesson-plan.md` for any day-specific materials)

Vocabulary introduced today

- *pratiyogi* (*modern coinage*) (IAST: *pratiyogi* (*modern coinage*); lit. “the number that adds to a base”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Draw a bar of length 100.** Cut a section of length 37 from the right end. Label the remaining bar.
2. **Decompose using 99.** Show that the bar of 100 = bar of 99 + bar of 1. The bar of 99 is easy to subtract from (every digit is 9).
3. **Visual mapping to digits.**
 - Top row: 99 partitioned into a section of 37 and a section of $(99-37) = 62$.
 - Bottom row: add the unit 1, splitting it into the rightmost cell.
 - Result: $(99-37)+1 = 100-37 = 63$.
4. **Generalise visually.** Show the same bar-decomposition for $1000-437$. The leading 9s allow digit-wise subtraction without borrow; the +1 at the end bumps the last digit.
5. **Independent task.** Each student draws their own bar-model proof for $1000 - 248 = 752$ and submits as proof of understanding.

Activity (15 min)

A scaled version of *Complement Flashcards* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Bar Model Proof.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 7 — Two’s Complement in Computers

Module: Subtraction — Nikhilam (All from 9) · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: show how the same idea underlies binary computer subtraction.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *Nikhilam Navataścaramaṁ Daśataḥ* (IAST: Nikhilam Navataścaramaṁ Daśataḥ; lit. “all from 9, the last from 10”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Bridge to computing.** Modern computers don’t have a subtraction circuit — they have an ADDER circuit. They subtract by adding the *two’s complement* of the subtrahend. This is the same Nikhilam idea, in binary.
2. **Binary Nikhilam.** For an 8-bit number:
 - One’s complement: flip every bit (= “all from 1,” the binary analogue of “all from 9”).
 - Two’s complement: one’s complement + 1.
3. **Worked example.** Compute $9 - 5$ in 8-bit binary:
 - $5 = 00000101$
 - Flip: 11111010 (one’s complement)
 - Add 1: 11111011 (two’s complement)
 - Add to 9 = 00001001 : $00001001 + 11111011 = 100000100$, drop the overflow bit → $00000100 = 4$. ✓
4. **Source.** D.E. Knuth, *The Art of Computer Programming*, vol. 2 — full treatment.
5. **Critical-thinking close.** “*The Nikhilam sūtra (or its decimal-complement principle) and binary two’s complement both date their origins to vastly different traditions and eras.*”

What does this tell us about (a) the universality of certain algorithmic ideas and (b) the limits of cultural-priority claims?”

Activity (15 min)

A scaled version of *Complement Flashcards* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Two’s Complement in Computers.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 8 — Speed Day

Module: Subtraction — Nikhilaṃ (All from 9) · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: complete a 30-problem mental subtraction sprint.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See [lesson-plan.md](#) for any day-specific materials)

Vocabulary introduced today

- *pratiyogi* (*modern coinage*) (IAST: *pratiyogi* (*modern coinage*); lit. “the number that adds to a base”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **30-problem mental sprint.** Mix:
 - Pure Nikhilam (subtract from 10, 100, 1000, 10000) — 15 problems
 - Add-the-complement (general subtractions) — 10 problems
 - Edge cases (trailing zeros, single-digit) — 5 problems
2. **5-minute timer.** Silence. Pen on paper.
3. **Pair-mark.** Swap and check.
4. **Distribution analysis.** “*Where did errors cluster?*” Most likely: edge cases (trailing zeros), large complements (10000 case).
5. **Personal best.** Record score; set target for next sprint.

Activity (15 min)

A scaled version of *Complement Flashcards* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Speed Day.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 9 — Mixed Practice

Module: Subtraction — Nikhilam (All from 9) · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: solve a mixed problem set comparing complement vs traditional borrow.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *Nikhilam Navataścaramaṁ Daśataḥ* (IAST: Nikhilam Navataścaramaṁ Daśataḥ; lit. “all from 9, the last from 10”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Mixed problem set.** 20 subtractions, classes mixed:
 - Simple borrows (use borrow or Nikhilam, your choice)
 - Multi-borrow (Nikhilam likely faster)
 - Subtraction from powers of 10 (Nikhilam dominates)
 - Add-the-complement (general)
2. **Strategy choice.** Each student picks the method per problem and writes the chosen method beside the answer.
3. **Cross-check.** Pairs swap; compare method choices. “*Did you both pick the same method? When did you disagree?*”
4. **Discussion.** “*Is Nikhilam ALWAYS better? When is plain borrowing fine?*” (Single-digit, small subtractions.)
5. **Method-comparison artefact.** Each student starts a “personal mental-math toolkit” page summarising when to use each method.

Activity (15 min)

A scaled version of *Complement Flashcards* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Mixed Practice.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 10 — Final Assessment

Module: Subtraction — Nikhilam (All from 9) · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: take the summative quiz.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See [lesson-plan.md](#) for any day-specific materials)

Vocabulary introduced today

- *pratiyogi* (*modern coinage*) (IAST: *pratiyogi* (*modern coinage*); lit. “the number that adds to a base”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Reflection essay (10 min).** *“In 200 words, explain Nikhilam to a younger student. Use one worked example and one limit-case.”*
2. **Summative quiz (25 min).** Computation, algebraic proof, two’s-complement connection. See `quizzes/summative.md`.
3. **Computer-subtraction explainer project share (10 min).** Each student presents their poster (see `assessments/project-brief.md`).
4. **Module close.** *“What’s one number-sense habit from this module you’ll keep using?”* Each student shares one sentence.
5. **Linkage.** Tomorrow Module 11 builds the multiplication side of Nikhilam — same sūtra, different application.

Activity (15 min)

A scaled version of *Complement Flashcards* (see `activities/`). Today’s slice: focus on the part of the activity that reinforces Final Assessment.

Wrap-up (5 min)

- One-sentence exit ticket: *“What’s one thing you learned today?”*
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Writing Prompts

Writing Prompts — Subtraction — Nikhilam (All from 9) (Senior)

Use these for in-class writing, homework, or assessment. Each prompt is suitable for a 300-word response.

1. In 3–4 sentences, explain Subtraction — Nikhilaṃ (All from 9) to a friend who hasn't taken this module.
2. Choose one Sanskrit term from this module. Define it, then describe one observation from your own life that matches.
3. Compare Subtraction — Nikhilaṃ (All from 9) with a similar concept from another module or from your science textbook. What's similar? Different?
4. Identify ONE claim in this module that you'd want to test. How would you test it?
5. What was the most surprising thing in this module? Why did it surprise you?
6. Pick one source cited in this module. Write 2 sentences explaining why it matters.
7. What is one common misconception about Subtraction — Nikhilaṃ (All from 9)? Rebut it in 2–3 sentences.
8. Imagine you are teaching this module to a younger student. What would you say first?
9. Read one primary-source passage cited in this module. Translate or paraphrase it; comment on what it adds to your understanding.
10. Identify a contemporary debate about the IKS framework presented here (e.g., is Tirthaji's mathematics genuinely Vedic? are *doṣas* a valid framework today?). Take a position with two pieces of evidence.

How to use

- Pick **one** prompt per day for the writing-prompts homework.
- Or assign all of them across the module, alternating between in-class and home.
- Senior students may treat any prompt as a 500-word essay topic.

Homework

Homework — Subtraction — Nikhilaṃ (All from 9) (Senior)

Light daily tasks. Each takes 10–15 minutes.

1. **Day 1 — The Sūtra — All from 9, Last from 10** · Write 2 sentences about today's concept. (~10 min)
2. **Day 2 — Complement to 1000 and Beyond** · Write 2 sentences about today's concept. (~10 min)
3. **Day 3 — Subtract from a Power of 10** · Write 2 sentences about today's concept. (~10 min)
4. **Day 4 — General Subtraction via Complements** · Write 2 sentences about today's concept. (~10 min)
5. **Day 5 — Why It Works — The Algebra** · Write 2 sentences about today's concept. (~10 min)
6. **Day 6 — Bar Model Proof** · Write 2 sentences about today's concept. (~10 min)

7. **Day 7 — Two's Complement in Computers** · Write 2 sentences about today's concept. (~10 min)
8. **Day 8 — Speed Day** · Write 2 sentences about today's concept. (~10 min)
9. **Day 9 — Mixed Practice** · Write 2 sentences about today's concept. (~10 min)
10. **Day 10 — Final Assessment** · Write 2 sentences about today's concept. (~10 min)

Notes

- Homework is intended to consolidate what was learned in class, not introduce new content.
- If a child struggles, the parent should consult the teacher rather than push.
- Capstone project days (Module 15) may have lighter daily homework so students can focus on the project.